

GUSIP High-Level Design / Architecture

Gujarat Unified Surveillance Intelligence Platform — hybrid overlay: Model 1 CCTV registry and GIS (mandatory) + Model 2 unified viewing and selective analytics + Model 3 VMS federation/middleware. **Not Model 4** (no replacement of departmental VMS, no central 24×7 video store).

1. Problem

Gujarat's CCTV estate is departmental and heterogeneous. The force needs statewide watchlists, multi-camera tracking, and one control-room picture without ripping out existing NVRs.

2. Principles

1. Do not break source operations. Adapters are read-oriented.
2. Centralise intelligence, not video. Metadata, events, embeddings, alerts, short clips only.
3. Federation before replacement (RTSP, ONVIF Profile S/T, vendor SDKs, Sentinel evaluation API).
4. Zero-trust internally; every query audited; purpose required on Investigate.
5. Horizontal workers: ingest, inference, tracking, matching scale independently.

3. Logical architecture

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Departmental VMS / NVR / Sentinel
  | RTSP / ONVIF / vendor API / official live-feed API (read-only)
Ingestion / Adapter Layer
  | sampled frames + camera telemetry
Federation Middleware
  | JWT RBAC, protocol translation, Redis/Kafka event bus
Central Intelligence
  | wall · ANPR · track · watchlist · alerts · search · GIS
Data: PostgreSQL+PostGIS · Redis · MinIO
UI: React control room
```

Upload architecture.svg with this PDF if the form wants a diagram file.

4. Mapping to official models

Model	GUSIP realisation
1 Registry / GIS	Camera master, PostGIS, health, AMC, gap analysis

2 Unified viewing	Control-room wall, auto-focus alerts, Investigate, cases
3 Federation	Per-source adapters; existing VMS untouched
4 Central VMS + AI farm	Out of scope for this proposal

5. PoC vs production

Block	PoC	Production
API	FastAPI	Same, behind gateway
Bus	Redis Streams	Kafka / Redpanda + ACLs
GIS	PostgreSQL 16 + PostGIS	HA + replicas
Evidence	Fernet snapshots + MinIO	KMS envelope encryption
Auth	JWT four roles	Keycloak OIDC, MFA
AI	Simulator $\equiv$ event contract + Tesseract on Sentinel frames	YOLOv8/11 + ByteTrack + Indian ANPR on regional GPUs

6. Key flows

Watchlist hit: adapter samples $\rightarrow$ detect/OCR $\rightarrow$ bus $\rightarrow$ match (coalesce same camera+list) $\rightarrow$ WebSocket $\rightarrow$ wall auto-focus $\rightarrow$ ack + audit. Target < 8 s on PoC corridor.

Investigate: purpose required $\rightarrow$ events/plate route $\rightarrow$ GIS polyline $\rightarrow$ case export (role-gated).

Outage: one VMS unreachable $\rightarrow$ those cameras offline; others continue; auto-reconnect.

7. PoC scope

50 own/demo cameras (rtsp/onvif/vendor_api) across Gujarat cities; 31 official Sentinel cameras; stolen corridor Ahmedabad $\rightarrow$ Gandhinagar; RBAC, audit, case export; documented 80k scale-out. Live VAHAN/SARTHI/eGujCop/AFIS are stubs only.

Companion diagram: workflow.svg · Full narrative: docs/HLD.md and docs/architecture.md in the repository.